Impulse changes momentum

antocciphysics | TEACHER KEY | CPA

Learning target: Connect force over time to momentum change.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 16; 3 course days later.*
g=3 N/kg; force 21 N inward.

02. Explain a worked example (worked)
Area 12 N·s; N·s=(kg·m/s²) s=kg·m/s.

03. Look and explain (model)
4 N×3 s=12 N·s impulse in positive direction.

04. Build the solution (practice)
3 m/s right.

05. Use the idea (reason)
Impulse/momentum change equal. Peak force and duration need not be equal.

06. Choose a method (mixed) *Review: Unit 2, Day 7; 12 course days later.*
4 N/kg toward the center; divide the surface value by 36.

07. Exit check (exit)
10 N·s right; 5 m/s right.

08. STRETCH: Apply it another way (stretch)
Examples 3 N for 4 s and 12 N for 1 s; equal areas with different heights and widths.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 16; 3 course days later.*
v=5 m/s; a≈2.778 m/s²; T≈11.31 s. Speed constant, direction changing.

02. Explain a worked example (worked)
Area 12 N·s; N·s=(kg·m/s²) s=kg·m/s.

03. Represent and interpret (model)
4 N×3 s=12 N·s impulse in positive direction.

04. Solve and justify (practice)
J=.5(-14)=-7 N·s; F=-350 N.

05. Analyze the reasoning (reason)
Impulse/momentum change equal. Peak force and duration need not be equal.

06. Choose a method (mixed) *Review: Unit 2, Day 7; 12 course days later.*
Center distances 4 R and 6 R; altitudes h/R=3 and 5.

07. Exit check (exit)
J=.5(.20)(30)=3 N·s; Δv=5 m/s in force direction.

08. STRETCH: Apply it another way (stretch)
Impulse 2(-2-3)=-10 N·s; average force -100 N.